

KULIYYAH OF INFORMATION AND COMMUNICATION TECHNOLOGY (KICT)

BIIT 2305 – Web Application Development

Section 2

Group Project Part 1

Project Name: HabitForge

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Proposal Video: <https://youtu.be/pdfqA2Cp2K8>

1. Introduction

HabitForge is a web-based application developed using the Laravel PHP framework that revolutionizes personal habit formation through behavioral psychology principles. Unlike conventional habit tracking applications that passively monitor user activities, SHBPS introduces an innovative **behavioral enforcement mechanism** that combines reward incentives with penalty consequences to drive consistent habit adherence.

The application addresses a fundamental challenge in habit formation: **accountability**. Research in behavioral psychology demonstrates that individuals are more motivated to avoid losses than to achieve equivalent gains (Loss Aversion Theory). SHBPS leverages this psychological principle by implementing a structured penalty system where users lose accumulated points, break streaks, and receive warnings when they fail to complete their scheduled habits.

Built on Laravel's robust Model-View-Controller (MVC) architecture, the system utilizes Eloquent ORM for database interactions, Blade templating engine for dynamic user interfaces, and Laravel's built-in authentication system for secure user management. The application transforms habit building from a passive tracking activity into an **active self-discipline enforcement system** that holds users accountable for their commitments.

The target audience includes students seeking to maintain study routines, professionals aiming to build work-life balance habits, fitness enthusiasts tracking exercise regimens, and anyone committed to personal development through consistent behavioral change. SHBPS distinguishes itself in the market by being more than a reminder tool—it serves as a **digital accountability partner** that enforces consequences, thereby increasing the psychological stakes of habit adherence.

This web application embodies the concept of "**self-discipline through technology**" where the system acts as an impartial enforcer of user-defined rules, creating an environment where consistency is not merely encouraged but systematically enforced through gamification elements and penalty mechanisms.

2. The Objective of the Proposed Web Application

HabitForge is developed with the following objectives:

Primary Objectives:

1. **To enforce habit accountability through automated penalty mechanisms** that deduct points, reset streaks, and issue warnings when users fail to complete scheduled habits, thereby creating psychological consequences for non-adherence.
2. **To implement a complete CRUD-based habit management system** that allows users to create, track, modify, and delete personal habits with customizable parameters including frequency, time scheduling, and priority levels.
3. **To develop an integrated reward and penalty scoring system** that gamifies the habit-building process, making consistency visually measurable through points, streaks, badges, and performance graphs.
4. **To provide real-time progress visualization** through interactive dashboards that display habit completion rates, streak histories, penalty records, and performance trends using dynamic charts and progress indicators.

Secondary Objectives:

5. **To create a secure multi-user platform** using Laravel's authentication system where each user maintains a personalized habit portfolio with private data isolation and session management.
6. **To design an intuitive and responsive user interface** following human-computer interaction principles that minimizes cognitive load and maximizes user engagement through strategic color schemes and layout design.
7. **To implement a notification and warning system** that proactively alerts users about upcoming deadlines, missed habits, and streak risks before penalties are applied.
8. **To build a scalable MVC architecture** that separates business logic (Models), user interface (Views), and request handling (Controllers), ensuring maintainable and extensible code structure following software engineering best practices.
9. **To integrate social accountability features** including leaderboards that allow users to compare performance metrics with peers, fostering healthy competition and community-driven motivation.
10. **To document the complete software development lifecycle** from requirements analysis through implementation to deployment, demonstrating comprehensive understanding of Laravel framework components and web application development principles.

3. Features and Functionalities of HabitForge

The HabitForge web application incorporates comprehensive features organized into functional modules:

Module 1: User Management & Authentication

Feature	Description	Functionality
User Registration	New account creation	Users register with name, email, password; validated server-side; unique email enforcement
User Login/Logout	Session-based authentication	Laravel Breeze/Jetstream authentication; CSRF protection; remember-me functionality
Password Reset	Self-service recovery	Email verification; secure token-based password reset flow
Profile Management	Personal information updates	Edit name, email, avatar; change password; account preferences
Role-Based Access	User permission levels	Regular users for personal habits; future admin panel for system management

Module 2: Habit Management (CRUD Operations)

Feature	Description	Functionality
Create Habit	Define new habits	Form with: habit name, description, category (study/exercise/health/productivity/custom), frequency (daily/weekly/custom), target time/completion window, priority level (low/medium/high)
Read/View Habits	Display habit catalog	Index page with filterable/sortable habit list; detailed view showing history and statistics; color-coded priority indicators
Update Habit	Modify existing habits	Edit any habit parameter; preserve historical data after modification; track modification timestamps
Delete Habit	Remove unwanted habits	Soft delete with confirmation dialog; option to retain or remove associated logs; archive functionality for dormant habits

Habit Categories	Organize by type	Predefined categories (Study, Exercise, Health, Productivity, Spiritual/Religious, Other); custom category creation
Habit Scheduling	Time-based configuration	Set specific times, days of week, reminder notifications; recurring vs one-time habits

Module 3: Daily Check-in System

Feature	Description	Functionality
Mark Complete	Record habit done	One-click check-in button; timestamp recording; optional note/reflection field
Mark Missed	Record uncompleted habit	Manual miss report or auto-detection after deadline passes; reason selection (optional)
Bulk Check-in	Multiple habits at once	Dashboard overview showing all pending habits; quick toggle complete/miss for each
Check-in History	View past records	Calendar view; date range filtering; export capability
Streak Tracking	Consecutive completion count	Current streak display; longest streak record; streak freeze option (earned through points)

Module 4: Penalty Mechanism (Core Innovation)

Feature	Description	Functionality
Point Deduction	Lose points for misses	Configurable penalty: -10 points for daily miss, -25 for weekly target miss; escalating penalties for repeated misses
Streak Reset	Break continuity tracking	Automatic streak reset to zero upon miss; visual indicator of broken streak; recovery bonus for rebuilding
Warning System	Pre-penalty alerts	Notification 1 hour before deadline; "Streak at Risk" badge; grace period option (configurable)
Penalty History	Record of all penalties	Detailed log showing date, habit, penalty type, points lost; sortable/filterable history view

Penalty Levels	Escalating consequences	Level 1: -10 points; Level 2: -25 points + warning; Level 3: -50 points + temporary point-freeze; Level 4: -100 points + mandatory review
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Module 5: Reward & Gamification System

Feature	Description	Functionality
Point Accumulation	Earn points for completion	+10 points per daily completion; bonus multipliers for streaks (x1.5 for 7-day, x2 for 30-day, x3 for 90-day)
Badge System	Achievement recognition	"Getting Started" (7-day streak), "Consistent" (30-day), "Unstoppable" (90-day), "Perfect Week", "Habit Master" (all habits complete for month)
Level Progression	Progressive tiers	Bronze (0-500 pts), Silver (501-1500), Gold (1501-3000), Platinum (3001-5000), Diamond (5000+)
Milestone Rewards	Streak-based bonuses	Streak freeze tokens at 7 days; custom theme unlock at 30 days; profile badge at 90 days
Weekly Summary	Performance recap	Email/dashboard summary every Sunday; points earned/lost; habits completed vs missed; comparison to previous week

Module 6: Dashboard & Analytics

Feature	Description	Functionality
Main Dashboard	All-in-one overview	Welcome message with user's level; today's pending habits; current streak; points balance; quick check-in buttons
Progress Charts	Visual performance data	Line graph: points over time; bar chart: weekly completion rates; pie chart: habit category distribution; radar chart: category performance
Streak Calendar	Visual tracking	GitHub-style contribution heatmap; color intensity reflects completion consistency; hover tooltips with details
Performance Analytics	Insights and trends	Completion rate percentage; most/least consistent habits; best performing days/times; penalty frequency analysis

Export Reports	Data portability	Download performance data as PDF/CSV; share achievements to social media; printable progress reports
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Module 7: Social Features

Feature	Description	Functionality
Leaderboard	Compare with peers	Global and friends-only rankings; weekly and all-time views; points-based and streak-based leaderboards
Friend System	Connect with others	Send/receive friend requests; view friends' public stats; cheer/encourage feature
Group Challenges	Shared habit goals	Create/join themed challenges; collective goals; group leaderboard within challenge timeframe
Achievement Sharing	Social proof	Share milestone achievements; generate shareable progress cards; optional social media integration

Module 8: Notifications & Reminders

Feature	Description	Functionality
Push Notifications	Browser-based alerts	Deadline approaching reminder; miss notification; streak achievement alert; penalty warning
Email Notifications	Extended alerts	Daily digest (pending habits); weekly report; penalty notification; achievement celebration
In-App Notifications	System messages	Notification center; read/unread status; notification preferences (opt-in/out per type)
Custom Reminders	User-defined alerts	Multiple reminder times; custom messages; snooze functionality

Module 9: Security & Data Management

Feature	Description	Functionality
Data Encryption	Protect sensitive information	Password hashing (bcrypt); secure session management; HTTPS enforcement; SQL injection prevention via Eloquent ORM

Input Validation	Prevent malicious data	Server-side validation for all forms; XSS protection via Blade escaping; CSRF tokens on all forms
Data Backup	Habit data preservation	Database backup routines; soft delete implementation; data recovery options
Privacy Controls	User data management	Public/private profile toggle; data export (GDPR compliance); account deletion with data purge option

Summary Table for Quick Reference

Module	Key Features	Laravel Components Used
User Management	Register, Login, Profile	Auth, Middleware, Sessions
Habit CRUD	Create, Read, Update, Delete	Models, Controllers, Migrations
Check-in System	Daily tracking, Streaks	Eloquent ORM, Relationships
Penalty Mechanism	Point deduction, Warnings	Events, Listeners, Jobs
Rewards System	Points, Badges, Levels	Eloquent, Collections
Dashboard	Charts, Calendar, Stats	Blade, JavaScript Chart Library
Social	Leaderboard, Friends	Polymorphic Relations
Notifications	Email, In-app alerts	Notifications, Mailables
Security	Validation, Encryption	Form Requests, Middleware

4. ERD

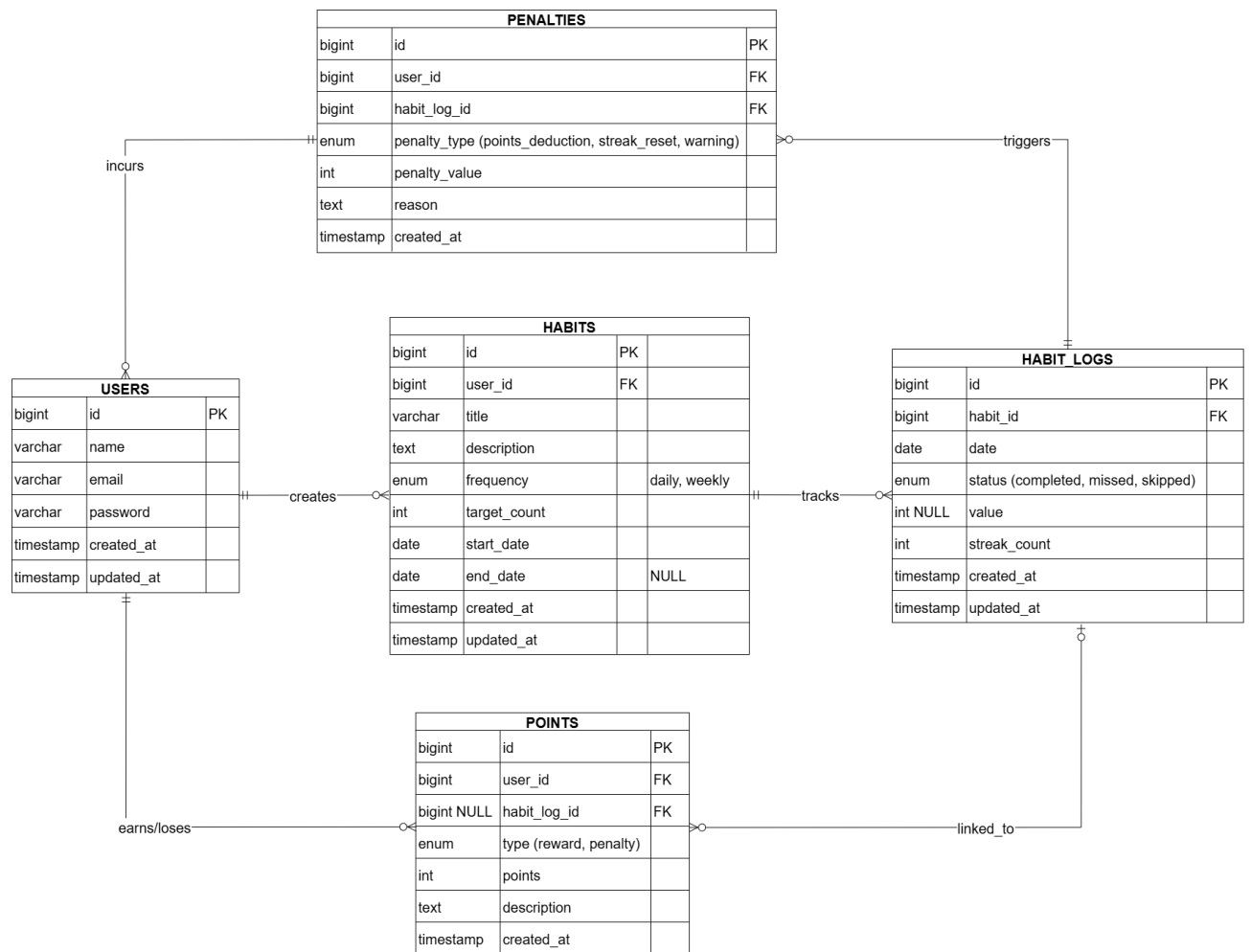


Table Definitions

1. Users Table

- **Purpose:** Stores user account information and authentication credentials.
- **Attributes:**
 - **id** (BIGINT, PK): Unique identifier for each user.
 - **name** (VARCHAR): Full name of the user.
 - **email** (VARCHAR): Unique email address for login.
 - **password** (VARCHAR): Encrypted password.
 - **created_at** / **updated_at** (TIMESTAMP): Standard Laravel timestamps for record tracking.

2. Habits Table

- **Purpose:** Stores the habits created by users, including their frequency and targets.
- **Attributes:**
 - **id** (BIGINT, PK): Unique identifier for each habit.

- **user_id** (BIGINT, FK): Links the habit to a specific user.
- **title** (VARCHAR): The name of the habit (e.g., "Daily Coding").
- **description** (TEXT): Detailed explanation of the habit.
- **frequency** (ENUM): Set as 'daily' or 'weekly'.
- **target_count** (INT): Number of times the habit should be performed.
- **start_date** (DATE): When the habit begins.
- **end_date** (DATE, NULL): Optional field for when the habit expires.

3. Habit_Logs Table

- **Purpose:** Tracks daily habit performance and maintains the user's streak.
- **Attributes:**
 - **id** (BIGINT, PK): Unique identifier for each log entry.
 - **habit_id** (BIGINT, FK): Links to the specific habit being logged.
 - **date** (DATE): The date the habit was performed or missed.
 - **status** (ENUM): Choices are 'completed', 'missed', or 'skipped'.
 - **streak_count** (INT): The current consecutive success count at the time of logging.

4. Penalties Table

- **Purpose:** Stores records of penalties applied when habits are missed.
- **Attributes:**
 - **id** (BIGINT, PK): Unique identifier for the penalty.
 - **user_id** (BIGINT, FK): Links the penalty to the user.
 - **habit_log_id** (BIGINT, FK): Links to the specific missed log entry.
 - **penalty_type** (ENUM): Includes 'points_deduction', 'streak_reset', or 'warning'.
 - **penalty_value** (INT): The numerical impact of the penalty.
 - **reason** (TEXT): Explanation of why the penalty was applied.

5. Points Table

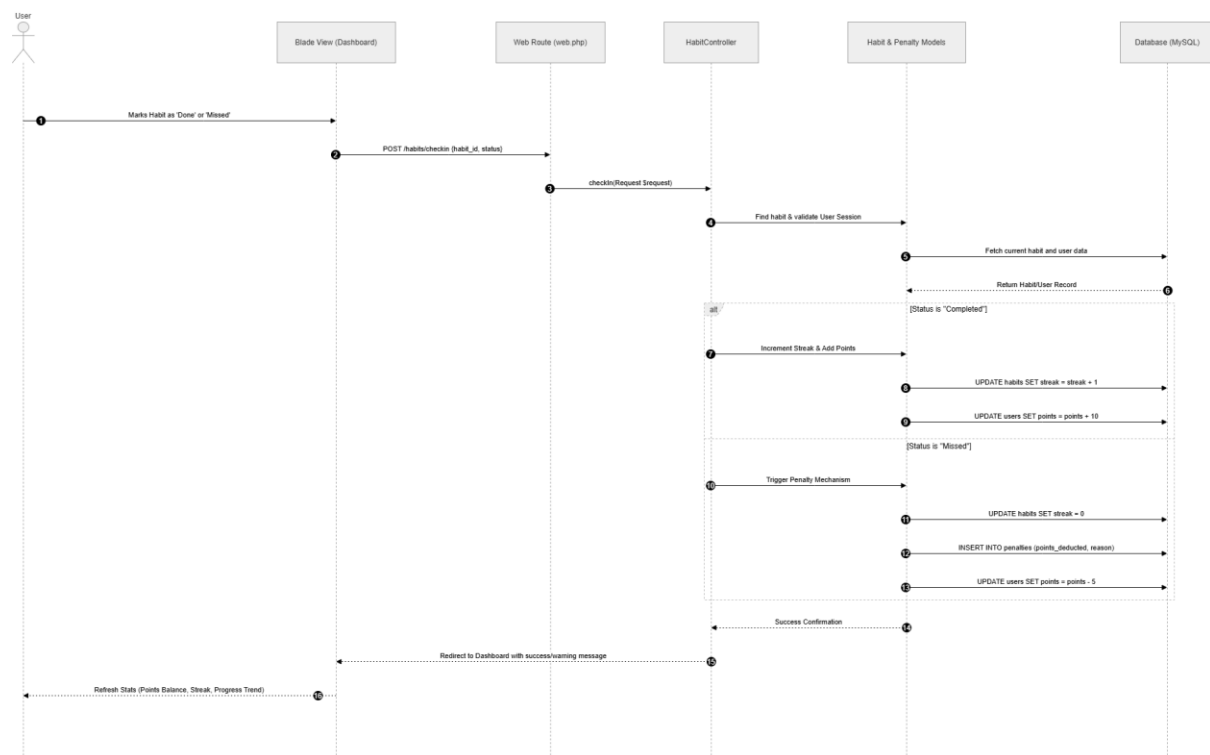
- **Purpose:** Tracks the history of points earned (rewards) or lost (penalties).
- **Attributes:**
 - **id** (BIGINT, PK): Unique identifier for the point transaction.
 - **user_id** (BIGINT, FK): Links the points to the user.
 - **habit_log_id** (BIGINT, FK, NULL): Optional link to a specific log entry.
 - **type** (ENUM): Categorized as 'reward' or 'penalty'.
 - **points** (INT): The number of points added or subtracted.

Entity Relationships Summary

To ensure the web application is functional, flexible, and reliable using the Laravel framework, the following relationships are established:

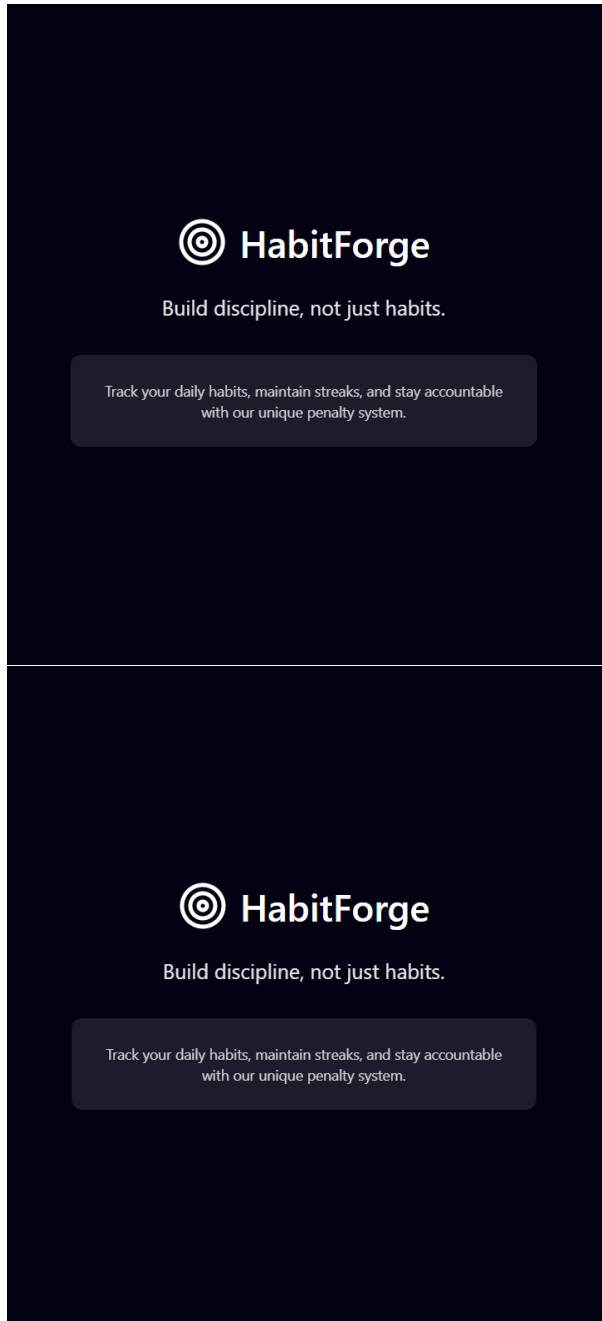
- **User to Habits (1:N):** A single user can create and manage multiple habits.
- **Habit to Habit_Logs (1:N):** Each habit has many log entries representing daily or weekly progress.
- **Habit_Log to Penalties (1:N):** A specific missed habit log can trigger one or more penalty records (e.g., a streak reset and a point deduction).
- **User to Penalties (1:N):** A user incurs many penalties over time for missed habits, allowing for a centralized penalty history.
- **User to Points (1:N):** A user has a history of many point transactions, which determines their overall standing or rank.
- **Habit_Log to Points (Optional):** Points can be linked to a habit log entry when a reward or penalty is directly tied to a specific day's performance.

5. Sequence Diagram



6. Mock-up

Link: <https://www.figma.com/board/Kli4PjkdNXFLIEIp87gPvO/mockup-lara?node-id=3-123&t=dochvFcYW5QTWFD8-0>



Login
Welcome back! Enter your credentials to continue.

Email

Password

☐ Remember Me [Forgot Password?](#)

Login

Don't have an account? [Register](#)

Create Account
Get started with building better habits.

Name

Email

Password

Register

Already have an account? [Login](#)

Welcome back, hazim

Track your habits and maintain your streak

Total Habits

0

Active habits

Current Streak

0

Days in a row

Points Balance

100

Total points

Weekly Penalties

0

Points deducted

Today's Habits

No habits yet. Create one to get started!

Create Habit

Progress Overview

Weekly Progress

Completion Rate

0%

Completed

0

Missed

0

My Habits

Manage your daily and weekly habits

+ Add New Habit

No habits yet. Create your first habit to get started!

Create New Habit

Add a new habit to track

Habit Name *

e.g., Morning Workout

Description

Optional description

Frequency

Daily

Time

09:00 AM

Penalty Points

5

Points deducted when this habit is missed

Active

Cancel

Create

Progress Overview

Track your habit completion over time

Total Completed



0

Habits completed

Total Missed



0

Habits missed

Success Rate



0%

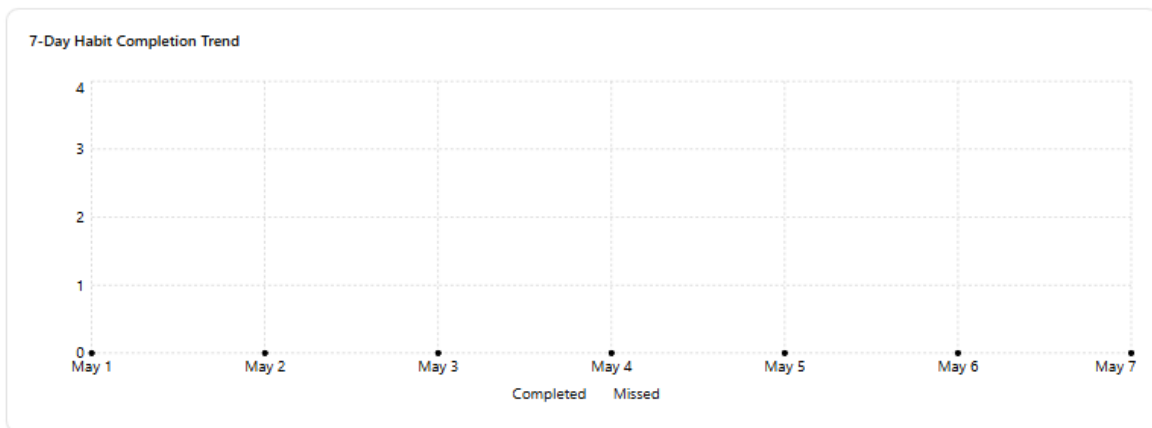
Overall completion

Longest Streak



0

Days in a row



Streak Tracker

Longest Streak

0



Best consecutive days

Current Streak

0



Start a new streak today!

Penalty History

Track missed habits and accountability

This Week



0

Penalties this week

Weekly Points Lost



-0

Points deducted

Total Lost



-0

All-time penalties

Penalty History



No penalties yet. Keep up the good work!

 HabitForge

 Dashboard
  Habits
  Progress
  Penalties
  **Leaderboard**

hazim
100 points
 

Leaderboard

See how you rank against other users

Your Ranking

#1

hazim
You

 100 pts
 0 day streak

Top Users




hazim
Rank #1



 100
 0

How Rankings Work

- Users are ranked primarily by total points
- In case of a tie, streak length determines the ranking
- Complete habits to earn points and climb the leaderboard
- Missing habits deducts penalty points and resets your streak

7. Reference

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